Construct-Aligned Disentangled Variational Autoencoders for Interpretable Buyer Personas: Measuring the Interpretability–Performance–Stability Frontier

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> **Provenance and honest reporting.** Phases 0–7 are complete. Every number below is recorded evidence from executed runs: baselines and sanity runs on the reference machine (RTX 4050), and the 288-run β×λ sweep on a Kaggle T4 at commit `d867713` (`git_dirty=false` , config hashes `973131b7` / `0a53edc0`), analysed by `run_phase6` . Reproduction was verified bit-exactly (`atol=1e-9`) on both datasets and again inside a fresh clone. Canonical copies of every table live in docs/NOTEBOOK.md; the `results/` tree is gitignored and regenerable from the committed configs and seeds. Nothing here is projected, estimated, or extrapolated. Analyses that are **implemented but not yet run** are labelled *pre-registered* and appear only in Section V-F — never as results.

***Abstract* — Buyer personas guide substantial marketing expenditure, yet the field's own survey of fifteen years of data-driven persona development names evaluation as an open problem. We reformulate persona quality as three separately measurable and separately falsifiable properties — downstream predictive lift on real behavioural labels, interpretability of latent axes with respect to named marketing constructs, and stability of persona assignments across seeds — and then measure what it costs to hold them at once. The instrument is a Construct-Aligned Disentangled Variational Autoencoder (CA-DVAE) whose latent vector is partitioned into a block trained by a linear auxiliary head to predict named constructs (recency-frequency-monetary value, price sensitivity, category affinity) and a free block carrying a β-weighted Kullback-Leibler term. Neither the architecture nor the aligned/free partition is claimed as novel; the contribution is the measurement reformulation, a leakage-audited evaluation harness, and the resulting frontier. Across two public datasets (2,240 customers; 2.55 million users aggregated from 110 million events), 288 sweep runs and six seeds per point, the findings are largely negative and specific. Non-neural incumbents set the bars: PCA reaches test PR-AUC 0.5677 ± 0.0729 on campaign response and three raw RFM features reach 0.1953 ± 0.0035 on future purchase, both beating a trained autoencoder. At its validation-selected operating point CA-DVAE reaches named-axis alignment R^2 up to 0.968 but sits significantly below both bars (0.5130 ± 0.0597 and 0.1798 ± 0.0051), and its personas are markedly less stable than incumbent personas (ARI 0.467 / 0.583 versus 0.975 / 0.962 for RFM+K-means). Three results survive as positive: against the *neural* incumbent the aligned model matches stability at matched performance and wins dormancy prediction in both test families; on the behavioural dataset the twelve named axes alone carry as much downstream signal as the autoencoder's full unnamed embedding; and the Mutual Information Gap is shown to be degenerate for model selection under correlated constructs, inverting with alignment strength. Interpretability in this setting is a purchase, not a free lunch, and this work prices it.**

***Index Terms* — Buyer personas, concept bottleneck models, customer segmentation, disentangled representation learning, evaluation methodology, interpretability, variational autoencoders.**

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I. Introduction

Customer segmentation into "buyer personas" is a standard instrument of marketing practice. Data-driven persona development has an active research literature of its own, but that literature's most comprehensive survey — 77 articles spanning 2005–2020 — lists **evaluation methods** among the field's persistent open gaps, alongside shared resources, standardization, and inclusivity [16]. Work addressing the gap exists: predictive personas have been created and validated by predictive accuracy on new customers [17]. What remains scarce is a protocol under which two persona *representations* can be compared on held-out future behaviour with the representation provably blind to the labels, repeated across seeds, and reported with the interpretability and stability those representations are actually chosen for.

We treat this as a measurement problem rather than an architecture problem. A persona representation is *good*, we argue, exactly to the extent that it satisfies three properties, each measurable and each able to fail independently:

1. **Downstream predictive lift.** A frozen persona representation should predict real future behaviour (campaign response, next purchase, dormancy) at least as well as incumbent representations, under a protocol that forbids the representation from ever seeing the labels.

2. **Interpretability.** Latent axes should align with *named* marketing constructs: recency-frequency-monetary value (RFM), price sensitivity, and category affinity, quantified by per-axis alignment R^2, the Mutual Information Gap (MIG), and the Separated Attribute Predictability (SAP) score.
3. **Stability.** Persona assignments should persist across random seeds and bootstrap resamples, quantified by the adjusted Rand index (ARI) and normalized mutual information (NMI).

The instrument for testing whether all three can be held simultaneously is a Construct-Aligned Disentangled VAE (CA-DVAE): a dense VAE over tabular customer features whose latent vector is partitioned into an aligned block, trained by a linear auxiliary head to predict the named constructs, and a free block carrying a β-weighted KL penalty. Sweeping the alignment weight λ and the disentanglement weight β traces a frontier over the three properties.

What we found. The result is largely negative, and the negative result is the contribution. Under leakage-safe evaluation the strongest baselines are the two representations the neural-segmentation literature most often treats as superseded: PCA on tabular survey data and three raw RFM features on behavioural data. At its validation-selected operating point CA-DVAE buys high named-axis alignment (R^2 up to 0.968) at a significant cost in predictive lift *and* a substantial cost in stability; incumbent personas are far more stable than aligned ones. Three findings run the other way and are reported with equal weight: against the *neural* incumbent specifically, the aligned model matches stability at matched performance and wins dormancy prediction in both test families; the twelve named axes alone carry as much downstream signal on the behavioural dataset as the autoencoder's full unnamed embedding, which rules out the "the names are decoration on a black box" reading; and MIG is degenerate as a selection criterion under correlated constructs.

The contributions of this work are as follows:

- A reformulation of persona quality as three measurable, falsifiable properties, with a concrete instrument for each and pre-registered bars fixed before any proposed-model run. Measuring persona quality by predictive accuracy is not new [17], nor is assessing segmentation stability by resampling agreement [28]; what is claimed is holding all three properties as co-equal criteria and measuring what each costs the others.
- A leakage-audited evaluation harness — frozen-representation protocol, temporal and user-disjoint splits, train-only construct fitting, six seeds per number, paired significance tests from two families with familywise correction, and validation-based hyperparameter selection — enforced by unit tests rather than by convention.
- An empirical interpretability–performance–stability frontier for construct-aligned persona representations, with the cost of interpretability quantified on both axes and the operating point selected on validation data only.
- Four findings that hold regardless of how the proposed model fares: learned neural embeddings do not automatically beat linear or heuristic incumbents under leakage-safe future-behaviour evaluation; hard cluster personas discard most of their parent embedding's predictive signal; stability tracks model simplicity and must never be read apart from performance, because near-collapsed models are trivially stable; and MIG inverts with alignment strength when constructs are correlated, making it unsafe for model selection in this setting.

Section II reviews related work and states what this work is not. Section III specifies the model, Section IV the data and constructs, Section V the protocol. Section VI reports results, Section VII discusses them, Sections VIII–IX cover validity threats and ethics, Section X reproducibility, and Section XI limitations and residual risk.

II. Related Work

A. Persona Construction and Evaluation

Data-driven persona development is surveyed comprehensively by Salminen *et al.* [16], who review 77 articles from 2005–2020 and identify evaluation methods as an open gap — the premise this work builds on, and the reason the claim here is "evaluation is underdeveloped", not "personas are never evaluated". Hsu *et al.* [17] are the nearest neighbour on the measurement axis, and the overlap is substantial enough to state plainly rather than minimize: their PP method builds personas with predictive data-mining algorithms and validates them by accuracy in predicting target customers, benchmarking predictive personas against traditional cluster-analysis personas using predictive accuracy as a single unified metric. **We therefore do not claim that measuring persona quality by predictive accuracy is new.** It is theirs.

What remains, and why the difference is not cosmetic. [17] resolves persona quality onto *one* metric. This paper's thesis is that a single metric is the problem: persona representations are chosen for interpretability and relied on for stability, and a one-axis criterion cannot express what is given up to obtain either. Three axes, and the measured trade-off between them, is the object of study — a frontier cannot exist in a one-axis framing. Secondarily, the unit of validation differs: [17] validates a persona *set* produced by a pipeline, whereas this work validates a *representation* by frozen transfer to labels it never saw, across seeds and multiple downstream tasks, under a leakage audit. The evaluation of embeddings by downstream transfer is itself routine in representation learning, including for customer embeddings specifically [29]; the novelty claimed here is its application to persona representations against incumbent *marketing* baselines, with leakage guards and multi-seed statistics.

Naming the tension is also not new. Boussebough *et al.* [30] balance performance against interpretability in multi-view customer segmentation, resolving it with a context-driven decision matrix over internal validation indices and business KPIs. They do not measure a frontier: there is no held-out downstream prediction and no stability analysis. The contribution claimed here is quantifying the trade-off, not observing that one exists.

B. Concept Supervision and Named Latent Axes

The construct-alignment head belongs to the concept-supervision family and is **not** claimed as a novel mechanism. Concept Bottleneck Models [18] predict human-specified concepts as an intermediate layer and then predict the label from those concepts alone. Concept whitening [19] modifies a normalization layer so that the axes of the latent space align with known concepts. Closest of all, CBM-AUC [20] places supervised concepts *and additional unsupervised concepts* in the same bottleneck, trained jointly — the same aligned/free partition used here. Partitioned supervised/unsupervised latent spaces recur across application fields, and the design should be read as conventional.

The distinction, in one sentence. A concept bottleneck routes the *task label* through the concepts and is trained on that label; CA-DVAE routes nothing through them — it is an unsupervised generative representation whose axes happen to be named, which is then frozen and transferred to downstream labels it never saw, so the interpretability and the predictive claim are measured on separate footings.

The failure mode this design inherits. Concept-supervised models with an unsupervised side channel are known to leak: learned concept representations encode information beyond the pre-defined concepts, and natural mitigations do not fully work [21]; concepts may not correspond to anything semantically meaningful in input space [22]. Two independent groups report this, which makes it a structural property of the design rather than a footnote. It bears directly on the headline interpretability evidence here: a high on-target alignment R^2 is the diagonal of a matrix whose off-diagonal must also be shown. Section V-F specifies the diagnostic we implemented in response, and Section VI-E reports the attribution evidence that bears on it.

In the customer domain specifically, Mancisidor *et al.* [23] steer a VAE's latent space with Weight of Evidence so the induced clustering reflects creditworthiness — prior art for "steer a customer VAE with a business quantity". The delta here is multiple named constructs held simultaneously, with the alignment strength swept rather than fixed.

C. Disentangled Representation Learning

The VAE [1] and β-VAE [2] form the base; FactorVAE [3] and the total-correlation decomposition of [4] penalize latent dependence directly. Kumar *et al.* [5] introduced SAP and Chen *et al.* [4] MIG, both adopted here. Locatello *et al.* [7] showed unsupervised disentanglement is impossible without inductive bias or supervision, and their follow-up [24] showed weak supervision suffices — so using explicit supervision, as here, is the field's standard escape rather than a differentiator. Importantly for this paper's hypothesis, Nai *et al.* [25] find that dimension-wise disentanglement is unnecessary for downstream performance and that *informativeness* is the better predictor, with prior positive findings explained by the correlation between the two. We use this to pre-state a directional hypothesis for the β sweep (Section V-E) rather than exploring blind.

D. Neural Customer Segmentation and Tabular Baselines

Deep Embedded Clustering [6] and autoencoder-plus-K-means pipelines dominate neural customer segmentation. Published evaluations typically report internal clustering indices on a single run; downstream predictive validation against the incumbent RFM representation [8] on held-out future behaviour, with seeds and significance tests, is rare. That evaluation gap, rather than any architectural gap, is this project's target. Our baseline findings are also consistent with a broader result: Grinsztajn *et al.* [26] show that tree-based models remain state of the art on medium-sized tabular data (~10k samples) even under matched tuning budgets, attributing the gap to neural sensitivity to uninformative features, rotation non-invariance, and difficulty fitting irregular functions. Dataset A ($n = 2{,}240$) sits squarely in that regime, so PCA and RFM beating a trained autoencoder is the *predicted* outcome — this work contributes the demonstration that the prediction extends to representation learning for segmentation under a frozen-transfer protocol.

E. Stability of Segmentation Solutions

The stability axis is **not** a criterion this paper proposes; it is an established marketing criterion applied here to learned representations. Dolnicar and Leisch [28] assess segmentation solutions by repeating the analysis on bootstrap samples and measuring partition agreement with the Rand index adjusted for chance, using reproducibility to decide whether data contain natural segments, structure without segments, or no structure at all — and distinguishing *natural*, *reproducible*, and *constructive* segmentation accordingly. That is the instrument used here, with training seeds added to bootstrap resamples as a second source of variation, since a learned representation can be unstable for reasons a fixed clustering algorithm cannot be. ARI [9] and NMI [10] are the underlying partition-agreement measures, and von Luxburg [11] frames bootstrap stability as a model-selection criterion in the clustering literature.

Section VI-D reports a confound that this framing makes easy to miss and that bears directly on [28]'s taxonomy: a degenerate, near-collapsed representation is *trivially* reproducible, so high stability can indicate an absence of structure rather than the presence of natural segments. Stability alone therefore cannot select a model, which is why Section V-D defines stability at matched performance.

III. Proposed Method: CA-DVAE

A. Problem Formulation

Each customer is a feature vector $x \in \mathbb{R}^D$ aggregated from raw records (Section IV). An encoder produces a posterior $q(z|x) = \mathcal{N}(\mu(x), \sigma^2(x))$ over $z \in \mathbb{R}^d$ with $d = 16$; a dense decoder reconstructs $\hat{x}(z)$. The latent vector is partitioned as

$$z = [\, z_{\text{aligned}} \in \mathbb{R}^a \mid z_{\text{free}} \in \mathbb{R}^{d-a} \,]$$

where a resolves by rule to $\min(n_{\text{constructs}}, d - 4)$, keeping at least four free dimensions (Dataset A: $a = 10$; Dataset B: $a = 12$).

B. Construct-Alignment Head

A **linear** head $A : \mathbb{R}^a \to \mathbb{R}^C$ predicts the C standardized construct targets from the aligned block; the alignment loss is summed mean-squared error. Linearity is a deliberate design constraint, not a simplification: it forces each construct to correspond to an interpretable *direction* in latent space, so the per-axis alignment score and latent-traversal persona cards are well defined. A nonlinear head would reintroduce the black box the project exists to remove — and would also be the easiest way to close the lift gap reported in Section VI-B while hollowing out the claim, which is why it was rejected rather than left as future work.

C. Loss and Ablation Structure

$$\mathcal{L} = \underbrace{\|x - \hat{x}\|^2}_{\text{recon}} \;+\; \beta \cdot \mathrm{KL}_{\text{free}} \;+\; w_a \cdot \mathrm{KL}_{\text{aligned}} \;+\; \lambda \cdot \mathcal{L}_{\text{align}}$$

with the standard ELBO reduction (sum over dimensions, mean over batch), so β and λ of order one are meaningful. β weights only the free-block KL; the aligned block keeps a unit-Gaussian prior at fixed weight $w_a = 1$ so it stays stochastic rather than collapsing under alignment pressure. The two components are independently ablatable as pure configuration points, with no separate code paths: λ = 0 recovers a pure β-VAE (the aligned block is then dissolved), and β = 1 recovers a VAE with alignment but no extra disentanglement pressure. The frozen embedding used downstream is the posterior mean μ over all d dimensions, identical in treatment to the autoencoder baseline, so any lift difference is attributable to the objective rather than the read-out.

The model is a small dense network trained comfortably within 6 GB of VRAM. No architectural novelty is claimed, and per Section II-B no novelty is claimed for the aligned/free partition either.

IV. Datasets and Construct Targets

A. Dataset A: Customer Personality Analysis

2,240 customers × 29 columns of survey-style tabular data (Kaggle, CC0-1.0) [13]. Label: response to the final marketing campaign (positive rate 14.9%); the five earlier campaign flags are legitimate strictly-past features. Cleaning is fit on train only (median imputation, winsorization at train percentiles, junk-level collapse). Splits are stratified 60/20/20 on the label, redrawn per seed; the test split is 448 rows (≈ 67 positives), which is small enough that Section V-C's interval estimates matter. Final matrix: 2,240 × 34.

B. Dataset B: eCommerce Behavior (Multi-Category Store)

109,950,743 raw events (October–November 2019) from the REES46 multi-category store [12], processed with Polars in streaming mode (the ~110M-row frame is never materialized; peak RAM stays bounded on a 16 GB machine). Users with ≥ 5 events in the feature window form the universe: **2.55 M users × 27 features**, cached to Parquet once (~200 s) and reused by every experiment.

Temporal separation is absolute: features aggregate the window [Oct 1, Nov 22); labels come from [Nov 22, Nov 30]; a unit test asserts every feature event precedes every label event. A skeptical-numbers pass corrected our own initial rationale here: the purchase surge in this data is Nov 16–17 (inside the feature window), not Black Friday, so the label window is a routine, promotion-unconfounded target. Train/val/test are user-disjoint (60/20/20 by seed-salted hash), blocking identity leakage. Three labels: `purchased` (primary, 3.3% positive), `churned` (dormancy proxy, 70.8% positive), and `next_category` (multiclass, first label-window purchase category).

C. Construct Targets (Alignment Supervision)

Construct	Dataset A (10 targets)	Dataset B (17 targets)
RFM (3)	Recency; Σ purchase counts; Σ spend	recency_days; n_purchases; total purchase value
Price sensitivity (1)	deal reliance = deals / purchases	−z(mean viewed price), a price-tier proxy*
Category affinity	6 spend shares (sum to 1)	13 known top-level event shares

*Honest limitation: the event log has no discount field, so B's price-sensitivity target measures the price tier a user browses, not discount responsiveness.

All construct targets are computed from pre-standardization tables, split with the same per-seed partition as the features, and standardized on **train only**; a perturbation unit test proves val/test rows cannot move the fitted scaler.

A property that shapes the whole evaluation. The constructs are *deterministic functions of the engineered features* — recency, column sums, spend shares. The alignment head therefore predicts something already present in its own input. This matters twice: it means the alignment task carries no new information and can only shape the *geometry* of the latent space, and it means a representation with perfect named axes is available without any training at all, by using the constructs as coordinates. Section V-F pre-registers that baseline, and Section VI-E reports the attribution evidence that addresses the circularity directly.

V. Evaluation Protocol

A. Frozen-Representation Protocol

Every method, baseline or proposed, yields a per-user representation fit on train only, which is then **frozen**. Two downstream heads (L2-regularized logistic regression and a histogram gradient-boosted tree classifier) are trained on frozen train features and evaluated on the held-out test set. Downstream labels never touch representation learning. Metrics: ROC-AUC, PR-AUC (primary; both binary labels are imbalanced), tie-aware precision@k (deterministic and row-order invariant), and for the multiclass task top-k accuracy plus macro one-vs-rest AUC.

B. Baselines (All Implemented, Bar Fixed Before Any Proposed-Model Run)

RFM (continuous, 3 features), RFM + K-means, PCA at matched capacity (d = 16), autoencoder embedding (the designated hard baseline), AE + K-means, GMM on AE embeddings, and DEC [6]. The raw standardized feature vector is also run through the same heads as a reference **ceiling**, excluded from bar selection, answering the reviewer question "do you need a representation at all?".

Tuning parity. The proposed model receives 24 configurations per seed from the sweep; a single-configuration autoencoder would be an unfairly weak comparison in the proposed model's favour. Section VI-H reports a dedicated AE mini-sweep run to close that gap.

C. Statistical Protocol

Every headline number is mean ± std over **six seeds**, with paired significance against the strongest baseline reported from **both** test families (Wilcoxon signed-rank and paired t). Six seeds is not arbitrary: the two-sided Wilcoxon p-floor is $2^{-(n-1)}$, which equals 0.0625 at five seeds and cannot reach 0.05; at six seeds the floor is 0.03125, so both families can carry a claim.

Multiplicity, stated explicitly. That floor is also a hard constraint on what six seeds can support. The comparison family here is large — up to eight challengers × two heads × three tasks, and again per sweep grid point — and *every* Wilcoxon win in this paper sits exactly at the floor. Under Holm–Bonferroni correction [27] within a (task, head, metric) family, two floor-valued comparisons already give an adjusted p = 0.0625 and none survives at α = 0.05. We therefore report: (i) raw p-values from both families; (ii) Holm-adjusted values, implemented in `cadvae.eval.stats.holm_bonferroni`; and (iii) paired effect sizes, because at six seeds a p-value carries little information about magnitude while a marketing reader needs to know how much lift is lost. Claims in Section VI that rest on the Wilcoxon floor are marked as such and are supported by the paired t and the effect size, not by the Wilcoxon alone.

Seeds are not sampling units. On Dataset A the split is redrawn per seed, so seed variance mixes initialization variance with split variance and the paired test pairs across *different* test sets. `cadvae.eval.stats.bootstrap_ci` supplies percentile intervals over test rows, which measure a different thing — how much of a number is an accident of which users landed in the test split — and the two are never conflated in Section VI.

D. Interpretability and Stability Instruments

MIG [4] and SAP [5] follow the estimator conventions of [7], with one documented deviation: because our ground-truth factors (the construct targets) are continuous, both latents and factors are discretized by quantile (equal-mass) bins, which are robust to the heavy tails of monetary features. Ground-truthrecovery unit tests require a disentangled code to outscore a rotation-entangled code of identical information content.

Stability is pairwise cross-seed ARI/NMI of K-means personas on a fixed seed-independent user sample, plus bootstrap persistence [11]. Because a degenerate representation is trivially stable (Section VI-D), we also define and report **SMP — Stability at Matched Performance**: the maximum cross-seed ARI among configurations within the selection slack of the best validation downstream score. SMP excludes collapsed configurations by construction and is the honest form of the stability comparison.

E. Pre-Registered Hypotheses and Falsification Criteria

Fixed before any proposed-model run:

#	Hypothesis	Falsified if	Outcome
H1	A construct-aligned representation matches or beats the strongest baseline on primary downstream lift	Sweet-spot PR-AUC significantly below the bar in both test families	**Falsified** on both datasets (VI-B)
H2	Alignment produces axes that genuinely encode the named constructs	Best per-axis R^2 near zero, or no better than an unaligned control	**Supported** (VI-C)
H3	Construct anchoring stabilizes personas relative to incumbents	Incumbent personas at least as stable as aligned ones	**Falsified** (VI-D)
H4	Increasing β improves downstream lift	Lift flat or decreasing in β	**Falsified**, in the direction [25] predicts (VI-B)

Directional hypothesis for H4 was stated in advance from [25]: because informativeness predicts downstream performance better than dimension-wise disentanglement, increasing β should *cost* lift rather than buy it.

Is the negative result publishable? Yes, and this was decided before running: the bars are pre-registered, the protocol is leakage-audited, and "the interpretable model loses to three raw features under honest evaluation" is a result the applied literature needs. The project was designed so that its value does not depend on the effect existing.

F. Selection Hygiene and Pre-Registered Additions

An audit before any sweep compute found that the sweet-spot (β, λ) point would have been selected on test metrics, the classic tuned-on-test flaw. The sweep records validation metrics per run; selection is on validation, reporting on test, and a drift-guard test pins the dual-evaluation heads to the exact Phase-3 specification. The same audit added per-task best-baseline comparisons (avoiding straw-man significance tests) and atomic record writes.

Two analyses are **implemented and unit-tested but not yet run**, and are reported here as pre-registered protocol, never as results:

1. **Named-axis baselines** (`constructs`, `construct_pca` in `cadvae.eval.representations`). Because the constructs are deterministic functions of the features (Section IV-C), using them directly as coordinates gives a representation with alignment $R^2 = 1$ by construction, no training, and — with PCA of the construct residual appended — the same aligned/free structure CA-DVAE learns. This is the strip test for the alignment mechanism: if it matches CA-DVAE, the machinery is decoration. It is

eligible for the bar, and is given the benefit of the doubt where the construct count exceeds the latent budget (Dataset B: all 17 named axes retained rather than truncated to 16).

2. **Concept-leakage diagnostic** (`cadvae.eval.interpretability.leakage_diagnostic`). Following [21], [22], it reports for each construct the off-target R^2 of its winning axis (purity) and the construct's recoverability from the *free* block alone. A high on-target R^2 with a high free-block R^2 would mean the named axis is not where the information uniquely lives. Nineteen ground-truth tests verify that the diagnostic separates a clean partition from a leaky one whose on-target scores are identical.

VI. Results

All numbers are from committed, config-hashed runs with 6 seeds under the audited protocol. Baselines and repro checks ran on an RTX 4050 (6 GB); the sweep ran on a Kaggle T4 at commit `d867713`. Canonical tables are mirrored in docs/NOTEBOOK.md.

A. The Bars: Non-Neural Incumbents Win

TABLE I — Baseline bar, Dataset A, campaign response (6 seeds, mean ± std, base rate 0.149)

Representation	PR-AUC (logreg)	PR-AUC (gbt)	ROC-AUC (gbt)
raw features (ceiling)*	0.6148 ± 0.0347	0.6066 ± 0.0749	0.8929
PCA (d=16) — the bar	0.5613 ± 0.0349	**0.5677 ± 0.0729**	0.8746
Autoencoder (d=16)	0.5600 ± 0.0477	0.5156 ± 0.0682	0.8346
GMM on AE	0.3523 ± 0.0402	0.3973 ± 0.0615	0.7649
RFM (3 features)	0.3794 ± 0.0381	0.3396 ± 0.0462	0.7230
DEC	0.3304 ± 0.0436	0.3483 ± 0.0465	0.7080
AE + K-means	0.3384 ± 0.0524	0.3343 ± 0.0543	0.7315
RFM + K-means	0.2988 ± 0.0182	0.2988 ± 0.0182	0.7205

*Excluded from bar selection.

PCA significantly beats the trained autoencoder (paired t, $p = 0.0019$; Wilcoxon at the floor, $p = 0.03125$), and PCA is statistically indistinguishable from the raw ceiling ($p = 0.19$). At $n = 2{,}240$ linear compression loses nothing and the neural baseline is not the bar; PCA is. This *raised* the bar CA-DVAE must meet on Dataset A from 0.532 to 0.568. Section II-D notes this is the outcome [26] predicts for tabular data at this scale.

TABLE II — Baseline bar, Dataset B, future purchase (6 seeds, 200k train subsample, full 509k-user test set, base rate 0.033)

Representation	PR-AUC (logreg)	PR-AUC (gbt)	ROC-AUC (gbt)
raw features (ceiling)*	0.1743 ± 0.0032	0.2038 ± 0.0042	0.8015
RFM (3 features) — the bar	0.1792 ± 0.0024	**0.1953 ± 0.0035**	0.7892
Autoencoder (d=16)	0.1675 ± 0.0045	0.1723 ± 0.0021	0.7875
PCA (d=16)	0.1405 ± 0.0079	0.1649 ± 0.0073	0.7865
GMM on AE	0.0743 ± 0.0045	0.1148 ± 0.0048	0.7480
RFM + K-means	0.1073 ± 0.0064	0.1073 ± 0.0064	0.7220
DEC	0.0708 ± 0.0047	0.0819 ± 0.0051	0.7061
AE + K-means	0.0672 ± 0.0071	0.0672 ± 0.0071	0.6341

*Excluded from bar selection.

On behavioural data, three raw RFM features beat every learned representation. The raw ceiling sits significantly above RFM, so non-RFM features carry signal no learned representation captures. Hard one-hot cluster representations — the standard "persona" format — discard the most signal on both datasets, losing 40–65% of their parent embedding's PR-AUC.

TABLE III — Per-task best baselines, Dataset B (6 seeds)

Task	Metric	Best baseline	Value	RFM comparison
purchased (primary)	PR-AUC (gbt)	RFM	0.1953 ± 0.0035	—
churned	PR-AUC (gbt)	AE	0.8620 ± 0.0006	beats RFM 0.8498 every seed (t, $p = 1.7 \times 10^{-8}$)
next_category	acc@1	AE + K-means	0.5550 (stable across all 6 seeds)	RFM collapses to 0.4070, *below* the 0.5053 base rate ($p = 0.017$)

No single baseline wins all three tasks, and the per-task best baseline — not a single global bar — is what the proposed model is compared against, so that no comparison is made against a straw man. On `next_category`, AE + K-means at 0.5550 is the real bar, above raw features (0.4520, $p = 0.021$) and PCA ($p = 0.020$); several methods, RFM included, score *below* the 0.5053 base rate, because the heads optimize log-loss rather than top-1 accuracy. RFM's collapse is structural: it contains no category information at all (macro-OVR AUC 0.52 versus 0.63–0.66 for the AE family).

The plain AE embedding is **bimodal across seeds** on `next_category` (macro-OVR AUC 0.73 / 0.74 / 0.73 on seeds 0–2 versus 0.54 / 0.52 / 0.52 on seeds 3–5; acc@1 0.58 versus 0.41–0.47), meaning the unregularized AE sometimes fails to allocate latent capacity to category structure at all. This is the observation that originally motivated construct anchoring as a stability remedy — a motivation Section VI-D goes on to falsify.

B. The Frontier: What Interpretability Costs in Lift

The Phase-5 campaign is a 6 β × 4 λ × 6 seed grid, 144 runs per dataset, 288 total, all complete. The $\lambda = 0$ column is the β-VAE ablation; the $\beta = 1$ row is the alignment-only ablation. Operating points are selected on **validation** downstream score and reported on **test**.

TABLE IV — Validation-selected operating points versus the pre-registered bars

Dataset	Sweet spot (β, λ)	Test PR-AUC (gbt)	Bar	Gap	Wilcoxon	paired t
A	(0.25, 4)	0.5130 ± 0.0597	0.5677 (PCA)	−0.055	0.0312 (floor)	3.0×10^{-3}
B	(0.10, 1)	0.1798 ± 0.0051	0.1953 (RFM)	−0.016	0.0312 (floor)	3.4×10^{-5}

H1 is falsified on both datasets. The interpretability cost in predictive lift is real, quantified, and significant under the paired t; the Wilcoxon values sit at the n = 6 floor and would not survive familywise correction on their own (Section V-C), but the t-test and the effect sizes carry the claim, and the direction is consistent across every seed.

The frontier is smooth and legible rather than a cliff. On Dataset A a weak-alignment point (β = 0.5, λ = 0.25) reaches 0.5647 — essentially at the bar — but with RFM-axis R^2 of only 0.35: the lift is recoverable precisely by giving up the naming. On Dataset B the best point anywhere in the grid is 0.1827 ± 0.0018 at (β = 0.1, λ = 0), i.e. with the alignment switched off entirely, and it still does not reach the RFM bar.

H4 is falsified in the predicted direction. High-β rows (β ∈ {1, 2}) are the worst on both datasets, confirming the posterior-collapse diagnostic recorded at the Phase-4 gate (KL_free ≈ 0.02 at β = 1) and matching the expectation set from [25]: pressure toward dimension-wise independence costs informativeness, and informativeness is what the downstream head needs.

C. Interpretability: The Mechanism Works, and MIG Does Not

At the selected points the named axes genuinely encode the constructs: Dataset A reaches mean alignment R^2 0.854 with RFM-axis R^2 **0.968**; Dataset B reaches RFM-axis R^2 up to 0.88 at λ = 4. Persona cards built from latent traversals are visually consistent with the names: Dataset A's price-sensitivity axis moves +deals / +kidhome / +recency against −income / −premium spend, which is deal-reliance semantics; every Dataset B affinity axis moves its own category share as the dominant feature.

A metric finding, reported rather than hidden. MIG is **degenerate as a selection criterion in this setting**. On Dataset B, MIG is *highest* at λ = 0 (0.26–0.35) and *drops* with alignment (0.07–0.10 at λ = 4). The cause is structural: aligning 12 latent dimensions to 17 correlated construct targets necessarily shares construct information across latents, and MIG's top-1-minus-top-2 gap penalizes exactly that. Selecting on MIG chose a pure β-VAE with no named axes at all as Dataset B's "sweet spot", producing an ablation table in which the proposed model and its own ablation were identical rows. Selection was moved to mean alignment R^2, and a sensitivity analysis confirms the failure is not a slack artifact: MIG-selection picks the degenerate λ = 0 point at every slack tested, while R^2-selection is slack-invariant on A and moves gracefully along the frontier on B. Both curves are still emitted. MIG is low everywhere on Dataset A (0.02–0.05), which is itself a caution about its usefulness for continuous, correlated business constructs.

D. Stability: A Cost, Not a Benefit — and a Confound

TABLE V — Cross-seed persona stability (ARI, K = 8, identical protocol)

Representation	Dataset A	Dataset B
RFM + K-means	0.975 ± 0.011	0.962 ± 0.033
AE + K-means	0.703 ± 0.068	0.745 ± 0.037
CA-DVAE, stable region ($\lambda \in [0.25, 1]$, low β)	0.64–0.70	0.58–0.73
CA-DVAE at the selected point	0.467	0.583

H3 is falsified. Both incumbents produce more stable personas than the selected aligned model. RFM+K-means is near-perfectly stable — unsurprisingly, since it clusters three deterministic features, so stability was never RFM's weakness. Strong alignment actively destabilizes: $\lambda = 4$ gives 0.46–0.55 across all β on Dataset A and is erratic on B (0.29–0.75 with large standard deviations at low β). Moderate alignment ($\lambda \in [0.25, 1]$) at low β is the stable-and-performant region, and the effect is non-monotone: on Dataset A, $\lambda = 0.25$ yields ARI 0.644, *above* the β-VAE ablation's 0.541, while $\lambda = 4$ yields 0.467, *below* it. Single-point stability claims would be reviewer-fragile in either direction, which is why the full 24-point stability surface is reported rather than one number.

The confound that makes raw stability comparisons meaningless. The most stable grid points are the most collapsed ones: Dataset A at ($\beta = 1, \lambda = 0$) reaches ARI 0.775 and Dataset B at ($\beta = 2, \lambda = 0$) reaches 0.834 — both at the *worst* downstream PR-AUC in their grids. A near-constant assignment is trivially reproducible. Stability must therefore be read jointly with downstream performance, never alone. A K-sensitivity check over $K \in \{5, 8, 12\}$ generalizes the point: the ordering "linear incumbents (RFM/PCA, 0.73–0.99) ≫ neural methods (AE/CA-DVAE, 0.41–0.78)" holds at every K on both datasets, so **stability tracks model simplicity**.

SMP — the honest comparison. Restricting to configurations within the selection slack of the best validation score, both datasets select ($\beta = 0.05, \lambda = 1$), where cross-seed ARI is **0.697 on A versus 0.703 for AE+K-means, and 0.725 on B versus 0.745**. At matched performance, aligned personas are as stable as the neural incumbent's. They remain far less stable than the non-neural incumbent's.

E. Attribution: The Named Axes Are Not Decoration

Because the constructs are deterministic functions of the features (Section IV-C), the sharpest objection to the interpretability claim is circularity: the aligned block may be merely re-encoding its own supervision targets. Decomposing downstream performance by latent block addresses this directly.

TABLE VI — Downstream PR-AUC (gbt) by latent block, at the selected points

Dataset	Full latent	Aligned block alone	Free block alone	Reference
A (β=0.25, λ=4)	0.5127	0.4429 (10 dims)	0.4456 (6 dims)	raw RFM 0.3794
B (β=0.10, λ=1)	0.1798	**0.1722** (12 dims)	0.1447 (4 dims)	AE full 0.1723

On Dataset B, **the twelve named axes alone carry as much downstream signal as the autoencoder's entire unnamed embedding**. On Dataset A the named block alone beats raw RFM, and the two blocks are complementary rather than redundant — neither alone reaches the full latent. The named axes are therefore load-bearing, not labels attached to a black box.

Two caveats are recorded with this table. First, it does not settle *concept leakage* (Section V-F): showing that the aligned block is informative is not the same as showing that each named axis carries only its own construct, which is what the pre-registered purity diagnostic measures. Second, the block-restricted heads reproduce the sweep records bit-exactly on B (2.8×10^{-17}) but deviate by 5.3×10^{-3} on A — cross-hardware encoding noise (sweep embeddings from a Kaggle T4 under torch 2.10, re-encoded on the RTX 4050 under torch 2.6) amplified by A's 448-row test set. B's 509k-row pipeline absorbs it. The magnitude is measured rather than assumed.

F. Where the Aligned Model Wins

TABLE VII — Dataset B multi-task outcome at the selected point ($\beta = 0.10$, $\lambda = 1$)

Task	CA-DVAE	Best baseline	Outcome
purchased	0.1798 ± 0.0051	RFM 0.1953	**Loss** (W 0.0312, t 3.4×10^{-5})
churned (dormancy)	0.8632	AE 0.8620	**Win**, both families (W 0.0312, t 3.6×10^{-4})
next_category	0.45–0.50 acc@1	AE + K-means 0.5550	**Loss** (t significant; W 0.0625, one step above the floor)

The dormancy win is real, tight across seeds, and small. In business units, precision at the top 10% of the ranked list is 0.9312 versus 0.9300 for the AE — **about 12 additional dormant users identified per 10,000 targeted** (W 0.0312, t 7.8×10^{-3}). It is reported at exactly that size. The `next_category` result is a loss, not parity: the head-level bimodality observed in the baselines is not cured by construct anchoring.

Taken together with Section VI-D's SMP result, the defensible positive claim is narrow and specific: **against the neural persona pipeline — the thing this literature actually proposes — the construct-aligned model matches stability at matched performance, wins dormancy prediction, and adds named axes at no measured cost in the primary task relative to that pipeline.** Against the non-neural incumbents it loses on lift and stability alike.

G. Ablations

Both components are ablatable as configuration points. $\lambda = 0$ (β-VAE, no alignment) is the best-performing region on Dataset B and produces no named axes; $\beta = 1$ with alignment is dominated by low-β alignment on both datasets. The ablation therefore shows that the alignment component *costs* primary lift while supplying the interpretability the paper is about, and that the disentanglement component costs lift without a compensating measured benefit in this setting — consistent with [25]. Neither component is decoration in the strip-test sense: each changes the measured outcome, and each has its own ablation row.

H. Baseline Tuning Parity

An AE mini-sweep on Dataset A (6 configurations × 6 seeds) was run so that the hard baseline is not undertuned relative to the proposed model's 24 configurations per seed. **The pre-registered bar is defended:** the best AE under the tree head reaches 0.5519, still below PCA's 0.5677. For transparency, a tuned latent-32 AE — twice the CA-DVAE capacity — reaches 0.5731 on the logistic head, at parity with PCA and not significantly different in either family (Wilcoxon 0.22 / 0.44; t 0.17 / 0.86). The pre-registered bar (gbt head) is unchanged, and the raw ceiling of 0.615 remains above everything.

VII. Discussion

What the study establishes. Under a leakage-safe, frozen-representation, multi-seed protocol on held-out future behaviour, the two representations most often treated as superseded in the neural-segmentation literature — PCA on tabular survey data and three raw RFM features on behavioural data —

are the strongest baselines, and the fashionable pipelines (DEC, AE + K-means one-hots) are the weakest. Any evaluation reporting only silhouette scores on a single seed would never observe this. Hard cluster assignment is the principal signal destroyer: every K = 8 one-hot representation loses 40–65% of its parent embedding's PR-AUC.

Interpretability has a price, and this is what it is. Named axes at $R^2 \approx 0.97$ cost 0.055 PR-AUC against PCA on Dataset A and 0.016 against RFM on Dataset B, plus roughly 0.3 ARI of persona stability relative to the non-neural incumbent. The trade-off is three-way, not two-way, and the stability leg is the one the project's own design originally got backwards. A practitioner can now make the trade explicitly instead of assuming it away.

When each representation is the right choice. If the goal is ranking customers by purchase propensity, use RFM: it is more accurate, more stable, free, and needs no GPU. If the goal is a persona scheme that names its own axes, supports traversal-based persona cards, and expresses category structure that RFM cannot represent at all, the aligned model buys that for a measured and modest price — and does so while matching the neural persona pipeline it replaces. If the goal is maximum accuracy irrespective of interpretation, use the raw feature vector with a gradient-boosted head; the ceiling is above every representation tested.

A methodological caution for this subfield. MIG should not be used to select models when ground-truth factors are continuous and correlated, which is the normal case for business constructs. It inverts with alignment strength for a structural reason (Section VI-C), and selecting on it here would have produced a paper whose proposed model was silently identical to its own ablation. Similarly, stability must never be reported apart from performance: the most stable configurations in both grids are the most collapsed ones.

VIII. Threats to Validity

Construct validity. The constructs are deterministic functions of the features, so the alignment task adds no information and can only shape geometry (Section IV-C). The attribution analysis (VI-E) rules out the strongest form of the circularity objection but not concept leakage, whose diagnostic is pre-registered and unrun. Dataset B's price-sensitivity target measures browsed price tier, not discount responsiveness.

Internal validity. On Dataset A the split is redrawn per seed, so paired tests pair across different test sets and seed variance mixes two sources (Section V-C). The 448-row test split makes PR-AUC genuinely unstable; bootstrap intervals are implemented for this reason. Sweep embeddings were produced on different hardware from the re-encoding used in VI-E, with a measured deviation of 5.3×10^{-3} on A.

Statistical validity. Every Wilcoxon result sits at the $n = 6$ discrete floor and none would survive familywise correction alone; claims rest on the paired t, the effect size, and per-seed consistency. Extending to more seeds *after* observing these p-values was deliberately rejected as optional stopping, which would be a worse attack surface than the one it closes.

External validity. One feature window and one label window on Dataset B — a rolling-origin replication across two or three origins would close the "the result is the window" objection and is the single most valuable unrun experiment. Two datasets, one domain (retail e-commerce plus a survey-style customer table), one language, one time period. The tabular-scale regime of Dataset A is exactly where [26] predicts trees and linear methods to win, so the A results should not be read as a general claim about neural representation learning.

Conclusion validity. The dormancy win is statistically real and practically small (≈ 12 users per 10,000). It is not evidence that the method is broadly superior, and is not presented as such.

IX. Ethics and Responsible Use

Both datasets are public and were used under their stated terms; Dataset B is used under an attribution and non-redistribution posture, and neither raw files nor derived caches are redistributed here. No personal identifiers are used: Dataset B is keyed by pseudonymous user IDs, and all features are aggregates.

Buyer-persona representations are targeting instruments, and the interpretability this paper measures cuts both ways: axes named "price sensitivity" make a model easier to audit *and* easier to use for price discrimination against deal-reliant customers, who in Dataset A co-vary with lower income and the presence of children in the household. Any deployment should be assessed for differential treatment across protected or proxy attributes before use; nothing in this evaluation certifies fairness, and no fairness metric is reported. Personas derived from behavioural logs also generalize poorly to people whose behaviour is sparse or atypical — Dataset B's universe requires ≥ 5 events, excluding the least-active users entirely, which is a selection effect a deployed system would inherit.

X. Reproducibility and Engineering Rigor

- **Determinism.** Global seed control; deterministic cuDNN; `CUBLAS_WORKSPACE_CONFIG` pinned; DEC's nondeterministic CUDA path rewritten (matmul distance expansion, CPU permutation draws); single-threaded BLAS on every sklearn fit that feeds a representation; deterministic row order everywhere positional operations occur. Identity re-runs are bit-reproducible on the reference machine.
- **Verified reproduction.** `repro_check` re-runs a pinned seed and diffs every numeric metric against recorded records: **REPRODUCTION OK at `atol=1e-9` (bit-exact) on both datasets**, and again inside a **fresh clone** built from the lockfile (69 tests passing in the clone). The Definition-of-Done requirement is demonstrated, not assumed.
- **Leakage guards as unit tests.** Perturbation proofs that train-fitted transforms ignore val/test on both datasets; temporal-ordering assertions; user-disjointness; label-window exclusion; construct-scaler isolation; a drift guard pinning sweep evaluation heads to the Phase-3 specification. Suite: **69 tests passing** at the close of Phase 7, plus **19 new tests** added with the pre-registered analyses of Section V-F (verified passing; ruff clean).
- **Cross-version portability.** Polars is pinned at 1.42.1 because Dataset B's split is a seed-salted Polars hash of `user_id` and that hash is not stable across versions — an unpinned run would silently drift the splits relative to the recorded bars. scikit-learn is pinned at 1.9.0 for identical heads.
- **Config-driven provenance.** Every run writes its resolved Hydra config, config hash, git commit, dirty flag, and seed. Ablations are configuration points, not code branches. Sweep record writes are atomic and the sweep is resume-safe.
- **No silent fallbacks.** Requesting CUDA when it is unavailable raises rather than degrading to CPU. A live GPU matmul, not `is_available()`, gates the sweep — this caught a Kaggle P100 whose driver reported availability while its kernels crashed.
- **Documented decisions.** docs/DECISIONS.md (D-001 … D-038), docs/NOTEBOOK.md, docs/CHECKLIST.md.

Computational cost. The full study is small, which matters given that a three-feature baseline wins: Dataset A's 144-run sweep takes ≈ 20 min (≈ 8 s/run) and Dataset B's ≈ 7.7 h (≈ 190 s/run) on a single T4; the Dataset B feature cache is one ~200 s streaming pass over 110M events; baselines and Phase-6 analysis add well under an hour. Total compute for every number in this paper is under nine GPU-hours. RFM costs three column aggregations.

XI. Limitations and Future Work

Limitations. Dataset B's price-sensitivity construct is a price-tier proxy. The churn label is a short-horizon dormancy proxy on a two-month log. Dataset B training uses a 200k-user subsample per seed (evaluation always uses the full test split). Section VIII states the validity threats in full.

Residual novelty risk, stated plainly. A dedicated novelty search covering 2018–2026 and extending into the marketing and information-systems literature was run before submission; the audit trail is in `research/ledger.md`. Its results are already folded into Section II, and they cost this paper two claims:

- The aligned/free latent partition is prior art [20], and no novelty is claimed for it.
- Measuring persona quality by predictive accuracy is prior art [17], and no novelty is claimed for it.
- The stability instrument — bootstrap resampling with the adjusted Rand index — is established marketing methodology [28], applied here rather than proposed.
- Naming the interpretability/performance tension in segmentation is prior art [30].

What survives, on the evidence searched, is the three-axis reformulation with a measured frontier, the leakage-audited multi-seed protocol applied to persona representations, and two methodological findings for which no prior report was found: MIG's inversion under correlated constructs, and the collapse–stability confound.

Two checks remain open and should be closed before submission. First, **[17]'s full text could not be obtained** — the publisher page, ACM DL, and every aggregator tried returned 403, so its characterization above rests on abstract snippets that were consistent across independent sources but are not the paper itself. If it turns out to report a frozen, held-out, multi-seed evaluation, the protocol delta narrows further. Second, an automated framework for interpretable customer segmentation in financial services (*Int. J. Financial Studies* 13(4):243) is described in secondary sources as using "RFM-based interpretability benchmarks" and "interpretability alignment measures"; it could not be retrieved and may be closer to the construct-alignment idea than anything cited here.

Future work, in priority order. (1) Run the two pre-registered analyses of Section V-F — the named-axis baselines are the strip test this paper most needs, since a zero-training `[constructs | PCA(residual)]` representation has perfect named axes by construction, and the concept-leakage diagnostic completes the interpretability evidence. (2) Rolling-origin replication on Dataset B. (3) A third dataset from a different domain. (4) Significance testing on stability deltas, which are currently reported as point estimates with seed spread. Not planned: a nonlinear alignment head, which would close the lift gap by removing the property the paper measures.

XII. Conclusion

This work replaces an unmeasured marketing artifact with three falsifiable measurements, builds the harness to take them honestly, and reports what the measurements say. They say that construct-aligned personas cost predictive lift and cost stability, that the incumbents this literature dismisses are hard to beat under honest evaluation, and that the interpretability being purchased is real — the named axes carry as much downstream signal on behavioural data as an autoencoder's entire embedding. Of four pre-registered hypotheses, three were falsified, including one the project's own design expected to hold. The methodological findings — MIG's inversion under correlated constructs, and the collapse-stability confound that makes unqualified stability comparisons meaningless — are offered as cautions to anyone evaluating persona representations next.

Acknowledgment

The author thanks REES46 Marketing Platform for the public eCommerce behavior dataset [12] and the maintainers of the Customer Personality Analysis dataset [13]. Dataset B is used under an attribution and non-redistribution posture: raw files and derived caches are never redistributed with this repository, and readers obtain the data from the original Kaggle source.

AI disclosure (per IEEE policy on AI-generated content). Generative AI (Claude, Anthropic) was used substantively in this project: implementation of the codebase under the phased protocol in CLAUDE.md, drafting of documentation, literature retrieval and verification, and drafting and structuring of this manuscript. All research direction, consequential scientific decisions (recorded per-decision in docs/DECISIONS.md), gate sign-offs, and accountability for the content rest with the human author. All reported numbers were produced by executed code on real data and are reproducible from the committed configurations and seeds; none were generated by a language model.

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Appendix A: Reproduction Guide

Requires: Windows, an NVIDIA GPU (developed against an RTX 4050 laptop, 6 GB VRAM), Python 3.12.

```
python -m pip install uv          # or: winget install astral-sh.uv
python -m uv sync                 # creates .venv from uv.lock (exact pinned deps, CUDA 12.4 torch)
.venv\Scripts\python.exe -m cadvae.utils.device   # GPU check — must print cuda_available: true
.venv\Scripts\python.exe -m pytest                # full test suite (leakage guards included)
```

(If `uv run` fails with a trampoline error on Windows, invoke the venv interpreter directly as above; it is the same environment.)

Data. Datasets are **not** committed (see Acknowledgment for the license posture). See data/README.md for download instructions and the expected layout under `data/raw/`. Then build the caches:

```
# Dataset A cache is built on demand by prepare(); Dataset B needs one streaming pass (~4 min):
.venv\Scripts\python.exe -c "from omegaconf import OmegaConf; from cadvae.data.ecommerce import
cache_user_table; cache_user_table(OmegaConf.load('configs/data/ecommerce.yaml'))"
```

Pipeline, exact commands per phase. All runs are config-driven (Hydra); every run directory records the resolved config, its hash, the git commit, dirty flag, and seed. Seeds: `eval.seeds = [0..5]` (D-023). Dataset B always takes the D-020 protocol overrides shown below.

```
# Phase 3 — baselines (fixes the bar; multi-task records)
.venv\Scripts\python.exe -m cadvae.eval.run_baselines data=personality
.venv\Scripts\python.exe -m cadvae.eval.run_baselines data=ecommerce eval.train_subsample=200000
model.max_epochs=40

# Phase 4 — single-point sanity run (gate evidence, not a result)
.venv\Scripts\python.exe -m cadvae.eval.run_cadvae_sanity data=personality

# Phase 5 — the beta x lambda sweep (resumable; ~20 min for A, ~7.7 h for B on a T4)
powercfg /change standby-timeout-ac 0     # once, before an overnight run
.venv\Scripts\python.exe -m cadvae.eval.run_cadvae_sweep data=personality
.venv\Scripts\python.exe -m cadvae.eval.run_cadvae_sweep data=ecommerce eval.train_subsample=200000
model.max_epochs=40
# Cloud alternative (Kaggle T4 — the path these results used): see cloud/README.md

# Phase 6 — analysis, trade-off curve, stability, persona cards (no training)
.venv\Scripts\python.exe -m cadvae.eval.run_phase6 data=personality
.venv\Scripts\python.exe -m cadvae.eval.run_phase6 data=ecommerce eval.train_subsample=200000

# Phase 7 — reproduction check (re-runs one pinned seed, diffs vs recorded records)
.venv\Scripts\python.exe -m cadvae.eval.repro_check data=personality
.venv\Scripts\python.exe -m cadvae.eval.repro_check data=ecommerce eval.train_subsample=200000
model.max_epochs=40
```

Pre-registered additions (Section V-F), not yet run. The named-axis baselines are selected through the same `eval.methods` mechanism as every other baseline:

```
.venv\Scripts\python.exe -m cadvae.eval.run_baselines data=personality "eval.methods=
[raw,pca,rfm,constructs,construct_pca,ae]"
```

Definition of done. A fresh clone, following this guide, reproduces all reported numbers (mean ± std over 6 seeds), the trade-off curves, the stability analysis, and every figure, verified by `repro_check` (exit 0). Verified 2026-07-11 on both datasets and in a fresh clone. The pipeline is bit-reproducible on a single machine (D-028); use `repro.atol` for cross-machine tolerance.

Appendix B: Repository Layout

```
configs/          Hydra configs: root (eval/sweep/phase6/repro blocks), data/, model/
src/cadvae/
  data/           Polars preprocessing, leakage-safe splits, constructs (Phase 1–2)
  models/         ae.py, dec.py, cadvae.py (Phase 3–4)
  eval/           protocol, representations, metrics, stats, interpretability,
                  sweep + analysis runners
  viz/            figures.py — trade-off curve, stability bars, persona cards
  utils/          seeding (determinism), device (no silent CPU fallback), run logging
cloud/            Kaggle/Colab/HF sweep setup (Kaggle T4 path verified — used for the sweep)
data/             raw + interim + processed (gitignored; processed = Parquet caches)
research/         literature ledger, change register, decision log for the manuscript
results/          per-run evidence (gitignored; numbers recorded in docs/ + README)
tests/            pytest — leakage guards, protocol smoke, metric ground-truth, phase-6 e2e
docs/             DECISIONS.md, NOTEBOOK.md, CHECKLIST.md, data cards
```

Appendix C: Result Artifacts

Regenerated by `run_phase6` into `results/phase6/{personality,ecommerce}_analysis/` (gitignored; numbers mirrored in docs/NOTEBOOK.md):

Artifact	Contents	Paper reference
`analysis.json`	Aggregated sweep, Pareto frontier, sweet-spot selection, paired tests	VI-B
`tradeoff_r2.png` / `tradeoff_mig.png`	Interpretability–performance curves on both x-axes	VI-B, VI-C
`stability.png`, `stability_map.{json,png}`	Cross-seed ARI/NMI at the selected point and over all 24 grid points	VI-D
`frontier3.png`	Three-way frontier: downstream × alignment R^2 × cross-seed ARI	VI-B, VI-D
`persona_cards.png`	Latent traversals along each named axis, standardized deltas with raw-unit annotations	VI-C
`attribution.json`	Downstream performance by latent block (aligned / free / full)	VI-E
`smp.json`	Stability at Matched Performance	VI-D
`baseline_stability.json`	Cross-seed ARI for incumbent persona pipelines	VI-D
`stability_k_sensitivity.json`	Stability over K ∈	VI-D
`selection_sensitivity.json`	Sweet-spot robustness to the selection slack	VI-C
`churn_business.json`	Dormancy win in precision@10% and users per 10,000	VI-F
`ae_minisweep.json`	Baseline tuning-parity check	VI-H
`multitask.md`, `ablations.md`	Per-task and ablation tables	VI-F, VI-G

For the full research protocol and phased execution rules, see CLAUDE.md.